

LEON AP, Spain

WMO: 080550

Lat: 42.588N

Lon: 5.651W

Elev: 3006

StdP: 13.17

Time Zone: 1.00 (EUC)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 23.0	(c) 25.9	(d) 14.7	(e) 12.9	(f) 34.7	(g) 18.8	(h) 15.8	(i) 34.6	(j) 27.4	(k) 43.1	(l) 24.6	(m) 45.6	(n) 2.8	(o) 330	(p) 0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 27.6	(c) 90.2	(d) 62.8	(e) 87.3	(f) 62.0	(g) 84.3	(h) 61.5	(i) 65.7	(j) 82.9	(k) 64.3	(l) 81.6	(m) 63.0	(n) 79.8	(o) 8.4	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.9	86.0	70.5	58.4	81.5	68.7	57.1	77.7	68.0	32.3	82.6	31.1	82.0	30.0	80.0	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	20.5	17.5	16.7	96.3	3.5	1.8	14.2	97.6	12.1	98.7	10.1	99.7	7.6	101.1
			15.8	69.3	3.4	4.6	13.3	72.5	11.3	75.2	9.4	77.8	7.0	81.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	38.0	40.5	45.1	48.6	55.3	62.9	68.2	67.5	62.7	54.9	45.4	40.8
	DBStd	11.94	4.99	5.79	5.81	6.34	5.39	7.36	5.37	5.47	6.23	5.77	5.82	4.69
	HDD50	1394	373	269	172	101	12	1	0	0	0	22	158	286
	HDD65	4901	838	687	617	492	305	127	30	36	117	314	588	750
	CDD50	2328	0	2	20	59	176	388	563	542	382	175	20	1
	CDD65	359	0	0	0	0	3	64	129	113	48	2	0	0
	CDH74	5036	0	0	0	9	113	900	1787	1558	617	52	0	0
	CDH80	1730	0	0	0	0	17	305	651	561	193	3	0	0
(14) Wind	WSAvg	6.6	6.2	7.7	8.5	8.3	6.9	6.4	6.0	6.1	5.8	5.7	6.3	5.3
(15) Precipitation	PrecAvg	22.4	2.4	2.2	1.3	1.9	2.0	1.5	1.0	0.8	1.6	2.4	2.5	2.6
	PrecMax	31.5	10.0	7.4	6.1	5.5	4.5	4.7	4.3	3.3	5.6	6.3	6.5	9.5
	PrecMin	14.3	0.0	0.2	0.0	0.1	0.2	0.1	0.0	0.0	0.0	0.2	0.0	0.2
	PrecStd	4.5	2.0	1.8	1.2	1.5	1.1	1.1	1.0	0.9	1.6	1.5	1.7	2.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.7	62.1	70.7	75.1	82.3	92.2	92.8	93.3	90.9	78.3	65.2	58.9
		MCWB	45.0	47.6	50.9	52.5	57.7	62.5	62.7	63.6	63.0	55.2	53.1	48.3
	2%	DB	52.5	57.4	66.4	71.1	77.0	87.5	89.9	89.4	85.6	74.9	61.0	54.0
		MCWB	45.3	46.7	49.7	51.9	55.7	62.5	63.0	62.1	62.3	55.8	51.3	47.9
	5%	DB	49.6	53.5	62.6	66.6	73.3	83.4	87.1	86.0	80.7	71.2	57.5	51.3
		MCWB	43.9	45.5	48.9	51.6	55.4	61.3	61.8	61.6	60.1	55.5	50.5	46.5
	10%	DB	46.7	50.1	58.3	62.2	69.4	79.1	84.0	83.0	76.8	66.6	54.3	48.6
		MCWB	42.7	44.4	47.8	50.3	53.7	60.1	61.9	60.7	58.5	54.2	49.5	45.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.9	51.2	53.8	56.9	60.3	66.3	68.9	67.4	65.1	61.0	56.3	52.1
		MCDB	53.1	56.2	66.1	69.6	74.3	82.0	86.4	87.9	83.9	69.8	61.5	53.8
	2%	WB	47.7	48.7	51.7	54.5	58.5	64.7	66.3	65.2	62.8	58.7	53.8	49.3
		MCDB	50.2	54.2	62.5	65.5	72.2	81.7	82.5	83.7	80.7	68.5	58.4	51.7
	5%	WB	45.1	46.8	49.9	52.8	56.7	63.1	64.4	63.6	61.4	57.0	52.1	47.4
		MCDB	47.8	51.5	60.1	63.4	68.8	79.6	81.1	81.0	77.0	67.1	55.7	49.9
	10%	WB	43.1	45.1	48.0	51.1	55.2	61.2	63.0	62.0	60.0	55.4	50.2	45.6
		MCDB	46.1	49.7	56.3	61.4	66.2	76.5	79.7	78.2	73.6	63.6	53.8	48.0
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	16.1	20.6	20.3	23.1	24.7	27.6	28.0	24.4	19.5	15.0	14.3
		MCDBR	20.8	22.1	29.6	30.0	31.7	30.8	32.4	31.8	29.7	28.2	20.5	20.1
		MCWBR	14.3	13.6	16.2	14.3	14.2	10.9	10.8	11.1	11.6	12.9	12.8	13.9
	5% WB	MCDBR	14.4	18.7	23.7	24.8	24.8	26.4	27.5	27.0	25.9	21.2	16.0	13.4
		MCWBR	11.2	12.4	13.7	12.9	11.9	10.3	11.0	11.3	11.3	11.1	10.8	10.5
(40) Clear-Sky Solar Irradiance	taub		0.281	0.292	0.324	0.337	0.351	0.356	0.343	0.346	0.330	0.319	0.292	0.278
	taud		2.534	2.503	2.431	2.402	2.380	2.384	2.403	2.393	2.449	2.500	2.541	2.540
	Ebn at Noon		278	292	293	295	292	290	292	289	286	277	271	269
	Edh at Noon		22	26	32	36	37	37	36	36	31	26	22	20
(44) All-Sky Solar Radiation	RadAvg		548	852	1227	1616	1956	2218	2292	2027	1556	973	612	486
(45)	RadStd		68	99	167	180	166	122	107	106	80	109	90	68

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page